

EXPERIENCE REPORT **OPEN ACCESS**

Design and Implementation of a State-Wide Network for Near Real-Time Public Health Surveillance and Research: The Minnesota Electronic Health Record Consortium Experience

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ABSTRACT

Objective: The Minnesota Electronic Health Record Consortium (MNEHRC) was established during the early days of the COVID-19 pandemic to provide data for public health surveillance from the eleven largest health care systems in Minnesota.

Methods: This is a descriptive study of the Consortium, which is a federated network that implements best practices for governance and data infrastructure to support public health surveillance and clinical research. We conducted an analysis of the Consortium members, governance structure, infrastructure, and the characteristics of the patient population.

Results: The Consortium health systems collect information from 105 hospitals, 773 clinics, 100 emergency departments and 29 040 providers. Information about the health systems and the demographic and clinical characteristics of its 5 471 367 patients is provided, which represents more than 90% of the patients in Minnesota. This manuscript also details the MNEHRC governance structure, working groups, data use agreements and technical infrastructure. The Consortium has produced several studies with state-wide impact. One study, Health Trends Across Communities in Minnesota, is described in detail to illustrate aspects of this collaboration.

Conclusion: MNEHRC has been a successful collaboration and vital resource for public health surveillance in the state of Minnesota. Initially, the Consortium focused on surveillance related to COVID-19 infections and vaccinations but has recently expanded into other public health and chronic disease research.

1 | Introduction

In 2019, researchers from several large health systems in Minnesota began to discuss with the Minnesota Department of Health (MDH) the idea of combining information from

electronic health records (EHR) to study statewide trends in key chronic conditions and public health indicators. To increase the chances that health systems would support the initiative, the executive leadership at the largest health systems in the state were surveyed and results suggested that initial

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efforts should focus on the opioid epidemic, the suicide epidemic, and chronic conditions.

From September 2019 to March 2020, three small groups focused on each of these issues and comprising members from eight of the health systems and two organizations with expertise in multisystem collaboration met to discuss how data could be shared. Over the course of several meetings, important general principles were developed as foundations for collaboration: shared governance, protecting the security and privacy of individual data through use of a federated data architecture, and a focus on health equity.

The plan was for all stakeholders to come together in person for the first time for a full day meeting on March 13, 2020, to lay the groundwork for establishing the Minnesota EHR Consortium (MNEHRC) focused on the three conditions prioritized by health systems' leadership.

However, shortly before that date, we recognized that we were on the cusp of a public health emergency, and that the Consortium could address gaps in traditional public health surveillance to support the response to the COVID-19 pandemic. Within a short time, MDH recognized the potential for MNEHRC to contribute to a coordinated statewide response to the pandemic and offered to participate in planning and seeking funding for the Consortium for COVID-19 surveillance. We continued recruiting other major health systems, and eventually, the eleven largest systems in the state, which collectively provide care to more than 90% of Minnesotans, agreed to participate. Tribal health systems, some independent hospitals/clinics and federally qualified health centers (FQHCs) are not currently included.

The earliest information was exchanged using a simple data model to describe COVID-19 testing and positivity rates among children and adults with and without viral symptoms [1]. Data were reported on a dashboard that was updated weekly (later bi-weekly, then monthly) which substantially expanded the granularity and speed of acquiring information compared to traditional infectious disease surveillance.

As COVID-19 vaccines became available, MNEHRC established a linkage with state data on Medicaid insurance and an immunization registry, and began reporting on vaccine uptake and disparities by demographic subgroups and location [2]. MNEHRC was also able to report timely data on vaccine and booster efficacy [3]. Additional linkages to data on incarcerated and homeless populations led to reports on COVID-19 vaccination in these under-studied groups [4].

When periodic reporting on COVID-19 testing, cases, and vaccination became more routine, the Consortium began work on other conditions of public health importance. MNEHRC now reports on hospitalizations and emergency department visits in Hennepin County related to substance use disorders. Data are reported quarterly in a new dashboard showing trends from 2012 to the present [5].

Our team had members who participated in other distributed health data networks and we incorporated the best practices

from those into how we organized MNEHRC. Some MNEHRC members were members of the Health Care Systems Research Network (HCSRN) and the National Patient-Centered Clinical Research Network (PCORnet) [6, 7]. From HCSRN, we wanted our network to have a common data model like its Virtual Data Warehouse and we wanted to have a very practical and collaborative data use agreement (DUA). We looked to PCORnet for privacy preserving record linkage (PPRL) to understand which patients overlapped across sites so we did not double count patients in our analysis. We eventually decided to use the Observational Health Data Sciences and Informatics (OHDSI) [8] Observational medical outcomes partnership (OMOP) common data model because it was being increasingly adopted for large health research collaborations. We also examined successful regional data sharing network examples of the Colorado Health Observation Regional Data Service (CHORDS) [9] and the OneFlorida data trust [10]. Those networks were instructive for how we could organize, although CHORDS only covers the Denver metropolitan area and OneFlorida uses a centralized research data repository. We believe that MNEHRC has incorporated the best practices from these other health networks and represents a learning public health system (LPHS) infrastructure as defined by Tenenbaum: "...the LPHS aims to create a dynamic, data-driven ecosystem that continuously improves public health interventions and policies" [11].

The purpose of this manuscript is to describe the current governance, data and analytic infrastructure, and practices of MNEHRC. We will summarize lessons learned and how we overcame challenges that arose over the past 5 years. The paper will illustrate the Consortium capabilities by describing a current project that goes beyond COVID: Health Trends Across Communities in Minnesota (HTAC) to provide estimates of chronic disease burden.

2 | Materials and Methods

The mission of MNEHRC is to improve health by informing policy and practice through data-driven collaboration among members of Minnesota's health care community. The Consortium's vision is to be the leading research network collaborating across Minnesota to equitably improve health outcomes.

A few key administrative processes support MNEHRC including our DUA, grant and project funding processes, project contracting, invoicing and reporting to funders. MNEHRC was initially funded by MDH with a program award from the centers for disease control and prevention (CDC). The MNEHRC governance model was developed collaboratively over 3 years and continues to evolve (Figure 1). The Governance Board is represented by a voting member from each organization supplying patient data to MNEHRC which serves as the primary decision-making body for MNEHRC and meets at least monthly. The Governance Board works toward successfully achieving the mission and goals of this emerging collaboration. The Governance Board recognizes that the Consortium will evolve over time, and that part of their responsibility is to actively monitor for needed governance changes as the complexity of the collaboration increases. Other organizations, agencies, or individuals deemed valuable to achieving the mission or expanding the influence and use of

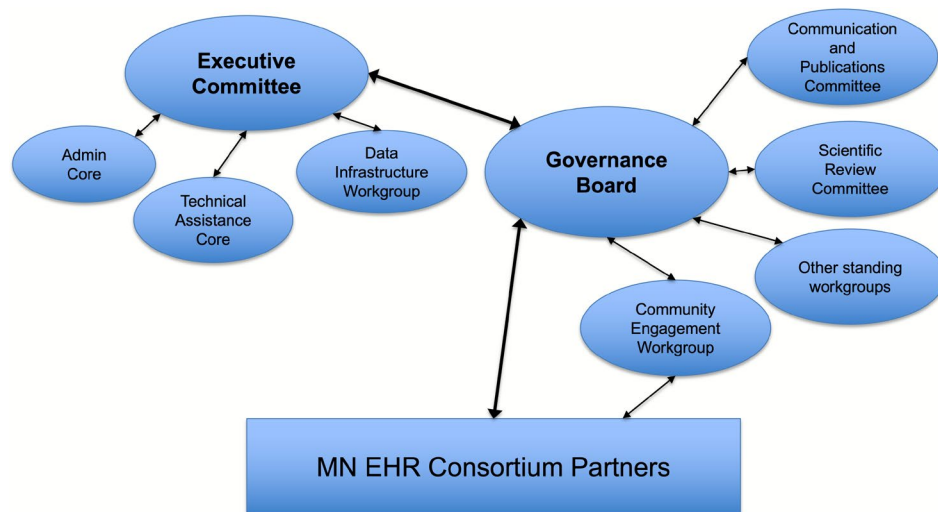


FIGURE 1 | MNEHRC governance model.

MNEHRC's work may be added at the Governance Board's discretion. The Governance Board seeks to use consensus as the primary decision-making strategy.

The Board votes on officers to serve on the Executive Committee. The Executive Committee is responsible for day-to-day Consortium operations and decisions, including data infrastructure, and preparing issues for discussion and voting by the Governance Board. Officers cycle through Presidential positions in one-year terms for a total of 3-years. Presidential positions are held by a person employed by a data-contributing health system. Two at-large positions consist of two-year terms and at least one shall be held by a person employed by a key funding partner (e.g., MDH).

The Scientific Review Committee is responsible for reviewing all project and manuscript proposals. Additional ad-hoc members may be added depending on the expertise needed for a thorough review. Projects and manuscripts approved by the Committee are sent to the Governance Board, which has the option to veto preliminary approvals by the Committee. In the absence of a veto, the approval is final.

The Administrative Core reports to the Executive Committee and carries out the day-to-day business of the MN EHR Consortium. Responsibilities include (1) coordinating Consortium meetings and maintaining meeting schedules, (2) maintaining an accurate list of contacts for study teams and administrative activities, (3) acting as the primary contact for administrative activities, including pre-award, post-award, and contracts, (4) conducting or coordinating outreach to new sites in partnership with the Executive Committee and other Consortium Partners, (5) providing oversight of contract and budget progress, (6) maintaining current DUAs and (7) developing templates for invoicing and maintaining tracking information to facilitate milestone invoicing by sites. The Administrative Core is also responsible for securing funding to support the Cores, the technical infrastructure, and effort from analysts at each site to manage their data and execute Consortium queries and tasks. MNEHRC was fortunate to be funded by MDH and CDC over the past 6 years. Each site has also contributed significant effort from site PIs, analysts and their respective database teams to ensure the Consortium's continued success.

The Technical Assistance Core reports to the Executive Committee and leads development and maintenance of MN EHR Consortium data models. Responsibilities include (1) conducting quality assurance, (2) maintaining shared data models, including project-specific intermediate files and specifications, (3) generating R Code for Consortium projects and streamlining transfer of project scripts, (4) maintaining Consortium deduplication procedures and other data management activities, (5) fielding questions from analysts and providing guidance and troubleshooting on data-related technical issues through regular meetings, (6) ensuring clear communication with technical aspects of each project, and (7) assisting in the development and maintenance of dashboards.

MNEHRC has established a Master DUA that outlines general roles for Consortium sites in projects. The DUA includes guidelines for analyses, deduplication, data receipt and aggregation, and suppression of cell sizes <11 in aggregate reports in order to prevent potential re-identification of individuals. The DUA also includes permitted uses (e.g., analyses, quality assurance work) and permitted disclosures (e.g., manuscripts, reports, white papers). Results will not be disclosed at the health system level beyond project-specific parties. Each new project requires a project specific appendix that includes information on participating health systems, funding, data elements, permitted uses and disclosures, and a data storage plan.

2.1 | Data Infrastructure

MNEHRC is implemented as a federated data network because it is important to our members that patient-level data never leaves a site. MNEHRC is a collaboration of health systems who compete in the health care market, but a federated approach allows sites to control their own data security, privacy and parameters of collaboration. The Consortium is not designed for nor does it allow health information exchange. Each site is responsible for maintaining their own database of EHR records using the OMOP common data model. [12] The Consortium spent 2 years implementing the OMOP model across all sites and worked together to achieve data standardization and harmonization by

mapping EHR concepts so that they mean the same across sites. Even though ten of the eleven sites used the Epic EHR, each site had differences in how they represented demographic and visit data. We worked to define standard mappings and each site implemented it in their system appropriately.

Figure 2 shows the high-level processes supporting our data and analytic capabilities.

There are a few processes that are centralized in the Data coordinating center (DCC) to provide privacy preserving record

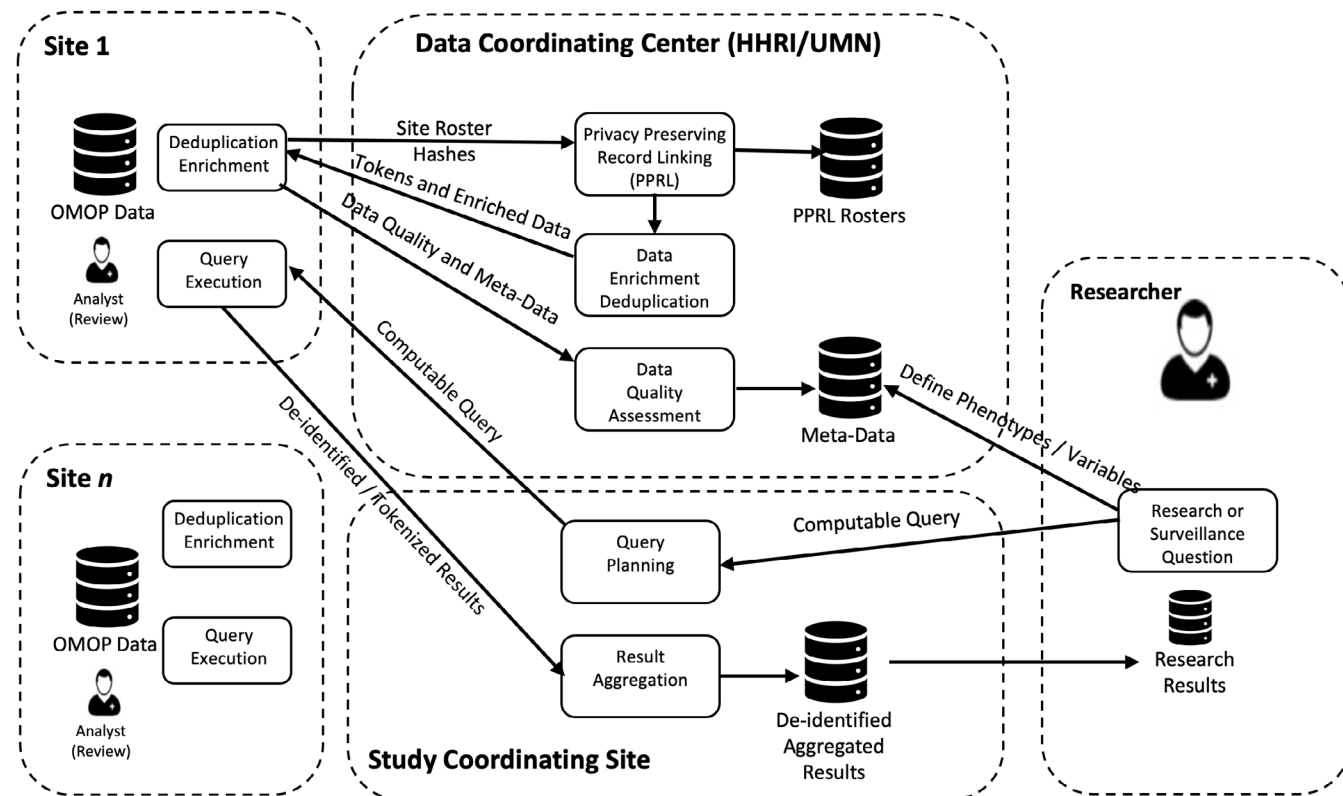


FIGURE 2 | MNEHRC data and analytic architecture.

TABLE 1 | Characteristics of MNEHRC member health care systems.

Site name	Number of hospitals	Number of clinics	Number of emergency departments	Number of providers*
Allina health	13	200	13	6500
CentraCare	9	30	9	1190
Children's hospitals and clinics of Minnesota	2	29	2	580
Essentia health	14	80	13	2830
HealthPartners	9	73	9	3100
Hennepin healthcare	1	8	1	950
M health fairview	11	91	9	3300
Mayo clinic rochester and mayo clinic health system	20	56	19	5950
North memorial health	2	12	2	440
Sanford health	22	172	22	2950
Minneapolis VA health care system	2	22	1	1250
Total:	105	773	100	29040

*Number of providers includes physicians and advanced practice providers.

linking for deduplication of patients, data enrichment and augmentation (i.e., Medicaid status, immunization history, homelessness, incarceration, death/mortality) and data quality assessment. MDH understood that these additional data sources are important for public health surveillance and worked with MNEHRC on DUAs to provide this information in a privacy preserving manner on a periodic basis (monthly and quarterly). The process for deduplication is critical to ensure accuracy and validity of study results. Since MNEHRC represents most of the health systems in Minnesota, there is significant overlap in the patient records across sites and we want to ensure that patients are not counted multiple times in a study. Records are linked in a privacy preserving manner using a secure one-way hash linking algorithm [13]. The algorithm performance was measured at OneFlorida and is sufficient for public health surveillance, providing a precision of 97% and a recall of 75%. Variables used for hash algorithms include a patient's first name, last name, date of birth, sex, phone number and zip code. Matching on the hash of these identifiers allows us to know that two records belong to the same person, but since the hash itself doesn't include personal health information, we do not know the identity of the patient. Seventy-five percent of patients covered by the MNEHRC have records in more than one health system and 32% have records in four or more systems.

We also receive a small set of health care utilization and demographic information about each patient from each site. This information is used to decide how patient information that may exist at more than one site is to be reported. These rules may differ depending on the study. For example, for COVID-19 surveillance, we prioritized data from sites that had the most recent visit data and most complete race/ethnicity data for the patient since that study was looking at potential disparities in infection and vaccination rates. The rules for which sites should report on which patients for which studies are encoded in a patient roster. The DCC returns patient roster data to the sites and augments the data with death information (from the state vital statistics database), vaccination status (for COVID-19 and influenza from the state immunization database), current Medicaid status, and homelessness and incarceration status. Using this architecture, any of the MNEHRC sites can initiate studies and any of the sites can serve as the study Coordinating Site aggregating results and performing the study analysis.

The DCC also provides data quality assessment and mitigation processes to ensure data integrity across the network. We use tools from OHDSI (Achilles, Data Quality Dashboard) in addition to our own custom scripts [8]. Data quality is assessed on a quarterly basis and the results are aggregated and stored longitudinally so

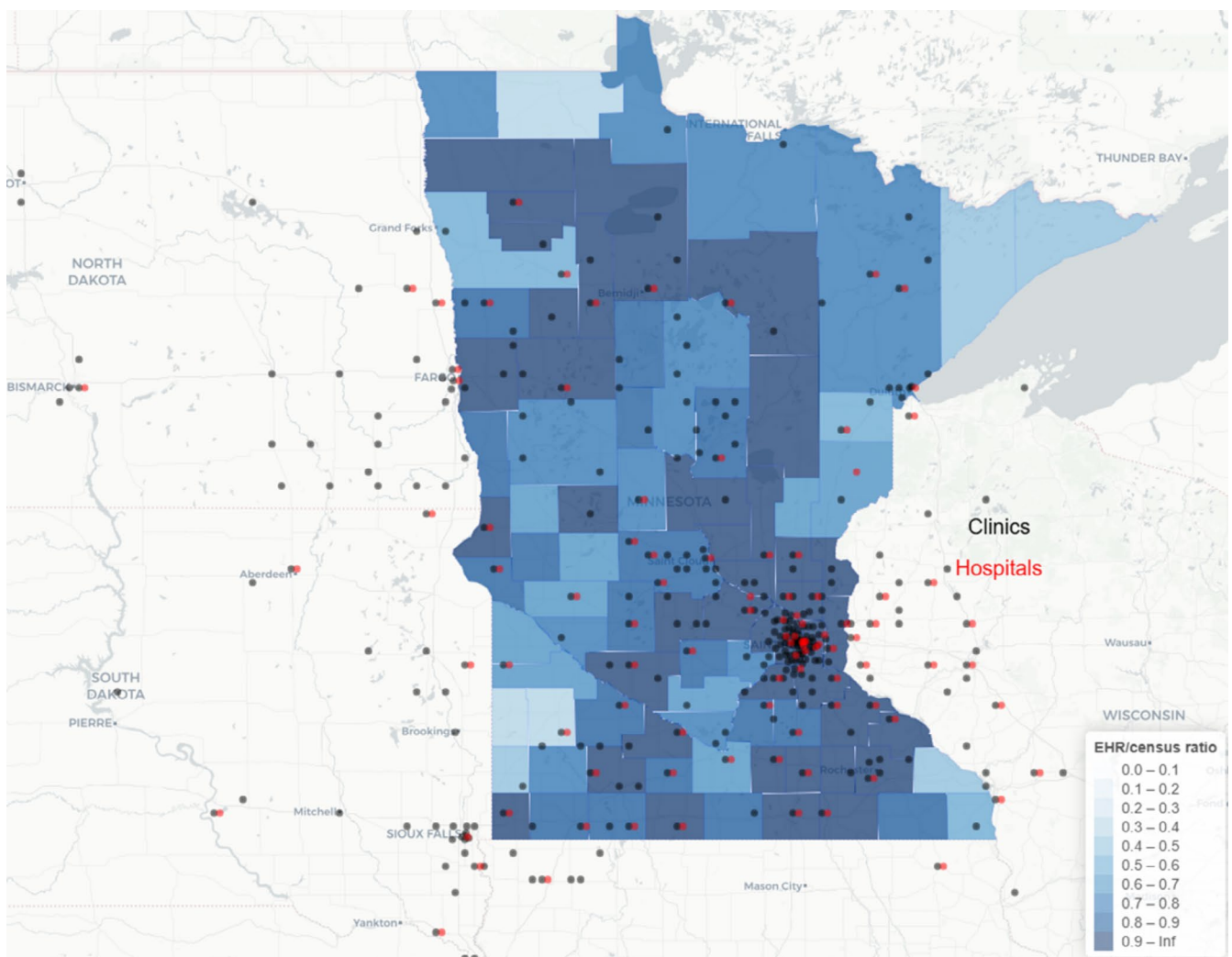


FIGURE 3 | MNEHRC coverage of Minnesota patient population.

that we can assess how data quality is changing over time and detect any anomalous site-level data changes. We have implemented data governance policies related to authentication, data transport and secure storage, retention and use of data policies.

2.2 | Analytic Processes

Our analytic process follows best practices [14] of (1) ensuring data quality fit for our study purpose, (2) codeset and variable definition using our common data model, (3) validating query scripts at one site, (4) distributing and executing scripts at all sites and (5) aggregating the results at the study Coordinating Site. To analyze data at each site and conduct studies or surveillance, we develop query scripts, written in the R programming language, that are distributed to each site to be run against the OMOP common data model. Most studies require that cohorts be defined using codesets that identify patient records that meet the criteria of the study (e.g., patients with Type 2 diabetes). Codesets are defined as lists of OMOP concept_ids that identify the condition, medication, measurement, and/or lab result of interest. The codesets and study scripts are checked into the Consortium version control system. Sites can then download the most current versions for a study, review them and then execute them in their secure environments. The aggregated results of the study query are then packaged, and sent to the study Coordinating Site for consolidation and analysis completion. Many studies are ongoing (e.g., the COVID-19 dashboard) and refresh the aggregated data at least every month. Any updates to the common data model or vocabularies are coordinated across all sites to occur at the same time. This architecture allows near real-time study data to be available. In practice, it takes time to run the scripts and collect the data at the Coordinating Site so a weekly cadence worked best (we were getting weekly updates for the COVID-19 surveillance).

3 | Results

Table 1 describes MNEHRC health system characteristics. Clinic and hospital locations for participating health systems are provided on a map (with one dot plotted on the map per zip code if multiple clinics or hospitals are located in the same zip code) in Figure 3. For each county in the state of MN, the coverage of the patient population (compared to 2020 census estimates) is provided, with darker shading indicating higher coverage. Based on the state of MN 2023 population estimate of 5.8 M [15], MNEHRC captures 94% of residents. Table 2 shows demographics and conditions for patients with at least one encounter in any MN health system from 2021 to 2023. Social vulnerability index and rurality use the Census Bureau definitions [16].

The MNEHRC data and analytic infrastructure has enabled the Consortium to undertake diverse studies and projects that have had state-wide impact including:

1. The Consortium has produced numerous studies detailing SARS-CoV-2 infection trends and vaccination uptake and efficacy across the state [1–4, 17].
2. The HTAC project provides prevalence estimates of over thirty health indicators for most Minnesotans, which has helped inform public health decisions [18].

TABLE 2 | MNEHRC Consortium Patient Population Characteristics.

Characteristic	Total number of patients
Total number of patients	5 471 367
Age at first visit, years, <i>n</i> (%)	
0–17	1 105 540 (20.2%)
18–44	1 953 810 (35.7%)
45–74	1 935 752 (35.4%)
75+	476 265 (8.7%)
Sex, <i>n</i> (%)	
Female	2 870 000 (52.5%)
Male	2 597 362 (47.5%)
Race, <i>n</i> (%)	
American Indian or Alaska native	56 926 (1.0%)
Asian	281 720 (5.1%)
Black or African American	503 333 (9.2%)
Hispanic	315 892 (5.8%)
White	3 802 333 (69.5%)
Other/Unknown/Missing	511 146 (9.3%)
Social vulnerability index (SVI) (%)	
Q1	23%
Q2	22%
Q3	25%
Q4	30%
Rural, (%)	39%
Health conditions, <i>n</i> (%)	
Hyperlipidemia	1 056 882 (19.3%)
Hypertension	1 031 210 (18.8%)
Depression	802 007 (14.7%)
Type 2 diabetes	388 704 (7.1%)
Asthma	378 255 (6.9%)
Ischemic heart disease	243 930 (4.5%)
Heart Failure	163 562 (3.0%)
Chronic obstructive pulmonary disease (COPD)	133 621 (2.4%)
Cancer (lung)	14 496 (0.3%)

3. Estimating prevalence of health conditions in homeless and incarcerated populations found notable disparities within those sub-populations, particularly among the distribution of substance use and psychiatric disorders [19].
4. MNEHRC executed a study that describes how outpatient use of telehealth expanded during the pandemic and found

that there was no change in the quality of care due to the use of telehealth visits [20].

5. The Consortium was able to quickly execute a study for MDH to characterize an outbreak of human parvovirus B19 infections among pregnant persons [21].

3.1 | Detailed Example: HTAC

HTAC provides an example of how the methods described above were implemented in a large ongoing project [18]. In mid-2022, MNEHRC collaborated with Bloomington Public Health and Hennepin County Public Health to propose building a state-wide surveillance system for chronic diseases and other key health indicators. A key component was to create a dashboard to display the prevalence of each indicator by age, sex, race, ethnicity, and census tract level geography. The proposal was selected by MDH for funding and started in the fall of 2022.

The study examined 31 conditions including chronic conditions, substance use disorders, mental health disorders, and maternal health. Several workgroups built codesets based initially on OMOP concept IDs for diagnosis codes, with a plan to later consider adding vital signs and laboratory results. Filters were applied to remove people who had died or did not reside in Minnesota. Indicators could be examined in special populations by merging data from those insured by Minnesota Medicaid, people experiencing homelessness, and people experiencing incarceration. The codesets were tested in several health systems, and then code was distributed to all health systems. Multiple

data iterations were required to obtain consistent results for central data aggregation. We used the data quality meta-data from Achilles and our own custom scripts to ensure that our code-sets covered all of the relevant data at each site. The dashboard was created, tested internally, and released to the public showing data for 2020–2023 in March 2024 (Figure 4) [22]. HTAC prevalence estimates provide researchers, clinicians, and policy-makers with easily accessible data to help inform the health of individuals in Minnesota. Local public health departments have also started using the data for their community health needs assessments [23].

4 | Discussion

MNEHRC has been a successful collaboration and vital resource for public health surveillance in the state of Minnesota. This effort informed policy and practice during the COVID-19 pandemic, and continues to be a valuable resource not only for COVID-19, influenza, and RSV surveillance, but for many other diseases of public health significance such as diabetes, hypertension, opioid use (including a substance use dashboard [5]), mental health, and chronic kidney disease. A major strength of the Consortium is the fact that the data represent over 90% of health care utilization throughout the state of Minnesota including representation of a large rural population. This depth of coverage affords a high level of ascertainment as well as a defined denominator for computing reliable estimates of disease occurrence. This percent coverage of the population is larger than other regional efforts such as OneFlorida (50% of the state covered) [10], Regenstrief [24] and CHORDs in Colorado

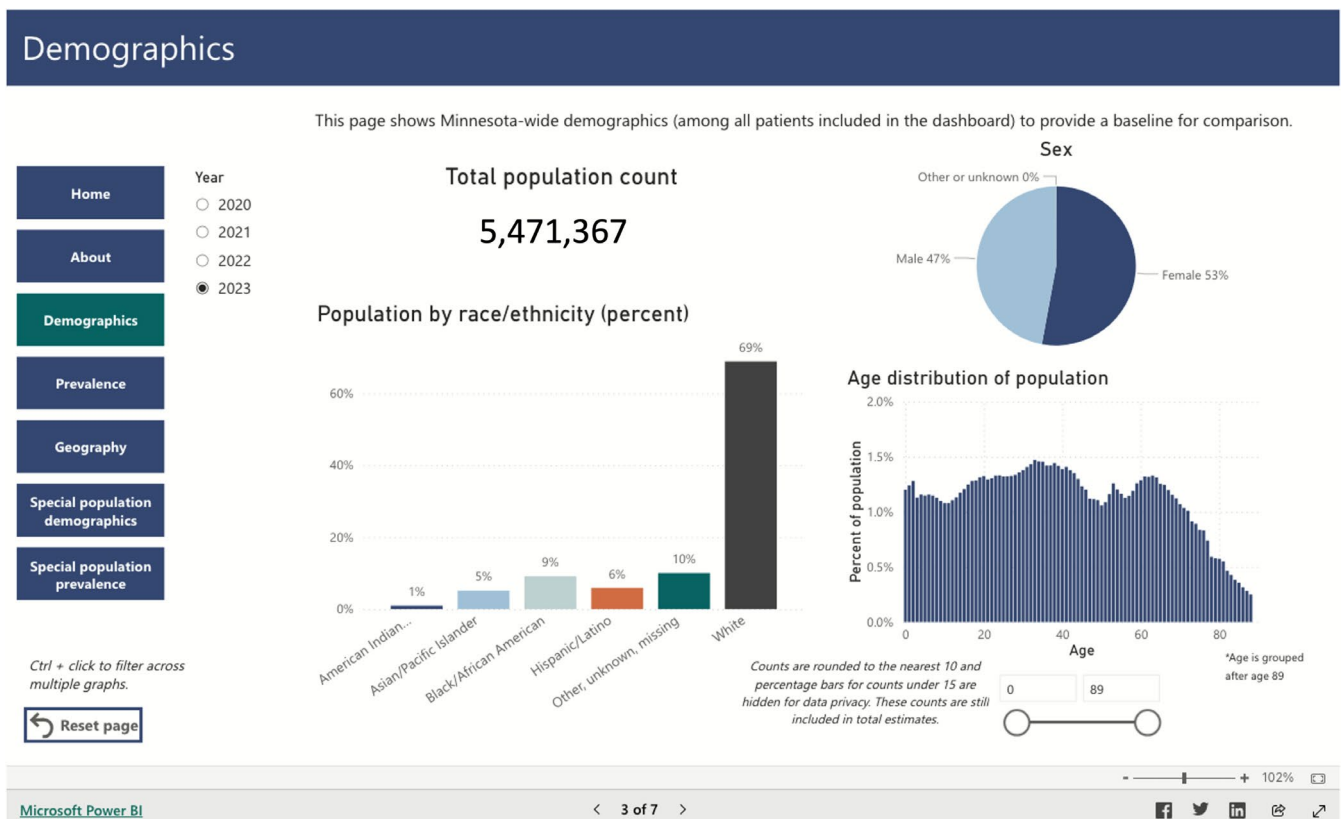


FIGURE 4 | HTAC dashboard.

(Denver metropolitan area only) [9]. In addition, MNEHRC represents an entire state so results can help inform public health policy at the state level. Recently, MNEHRC, the University of Minnesota and MDH were awarded grant funding to develop modeling and analytic infrastructure to obtain and analyze data to help decision makers weigh their options in a future public health emergency [25].

Some of the factors that were critical to our success are the knowledge and experience gained by learning from ongoing collaborations in other networks (PCORnet, HCSRN, etc.). We have a strong, collaborative partnership with MDH in identifying gaps and procuring funding. The Consortium membership includes many skillsets and capabilities, including clinical expertise in disease areas, public health, epidemiology, EHR, data infrastructure, data science, biostatistics, informatics, grant writing, and project management. Early on, we established an organizational structure and governance process that continues to evolve over time. Our initial DUA allowed us to start sharing data for COVID-19 surveillance, but we amended it later to allow analysis (but not publication) of datasets with small cell sizes. We incrementally developed our data models, technical support for sharing code, data governance, and analytic architecture and processes. The health systems have benefited from having an OMOP version of their data. Sites are able to provide OMOP data for their internal projects, research, and use OHDSI supported tools. Sites have also started to incorporate these data and processes into their own Learning Health System efforts by supporting internal metrics, reporting, and community health needs assessments.

There are several limitations with the current Consortium. While we have substantial data on racial/ethnic minority and rural patients who use our health systems, we are missing some very important patient populations (~10%) including patients served by most federally qualified health centers and other community clinics and several rural parts of the state. We are actively working to fill in our data gaps. Sustainable funding is also an issue that we are actively trying to address through applying for additional CDC and NIH grants as well as working with private foundations and charities. Our strategy is to allocate a portion of study funding for maintaining and enhancing our data and analytic infrastructure.

Acknowledgments

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Ethics Statement

The project was reviewed by the Institutional Review Boards of the data-contributing sites and was determined to be exempt from further review and approval in accordance with regulations governing public health surveillance.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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